SARDAR VALLABHBHAI
NATIONAL INSTITUTE OF TECHNOLOGY
SURAT

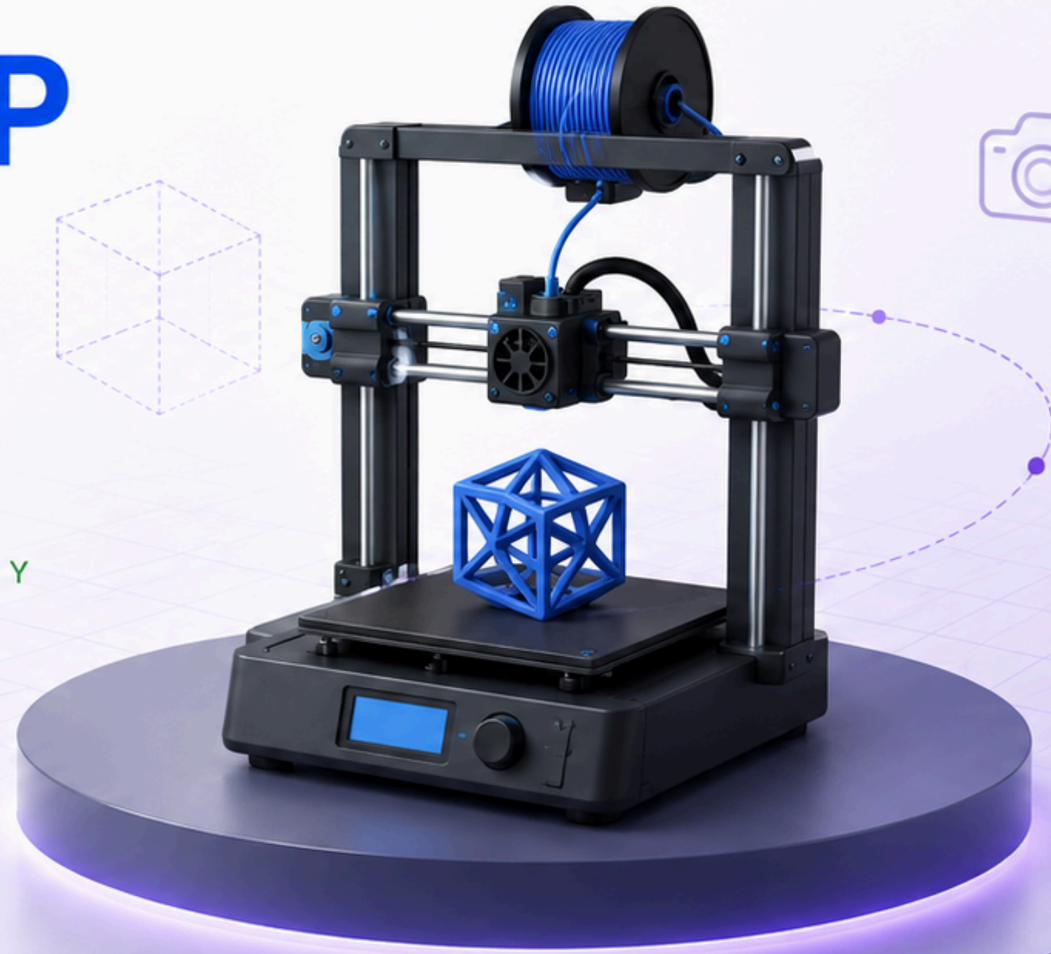
Spline
Interact • Visualize • Present

womp
Create • Modify • Refine

EXPLORING SPLINE AND WOMP FOR 3D VISION

Understanding 3D Representation, Visualization and Interaction through Practical Exploration

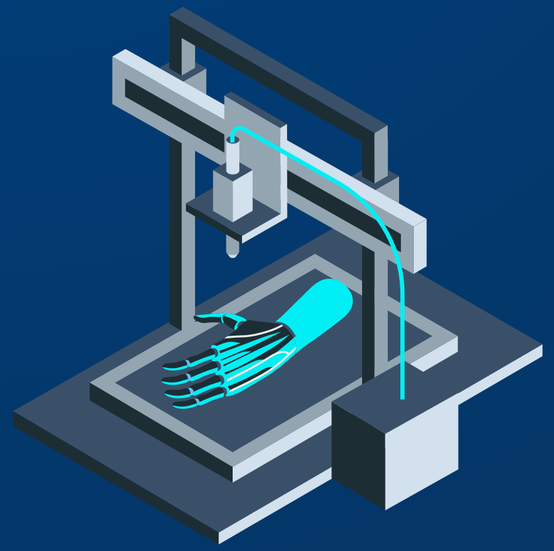
Presented by	:	Kishan Sahu
Roll No.	:	P26DS017
Subject	:	Computer Vision and Image Processing Lab
Course	:	M.Tech Data Science – 1st Sem
Faculty Guide	:	Prof. Mukesh Zaveri



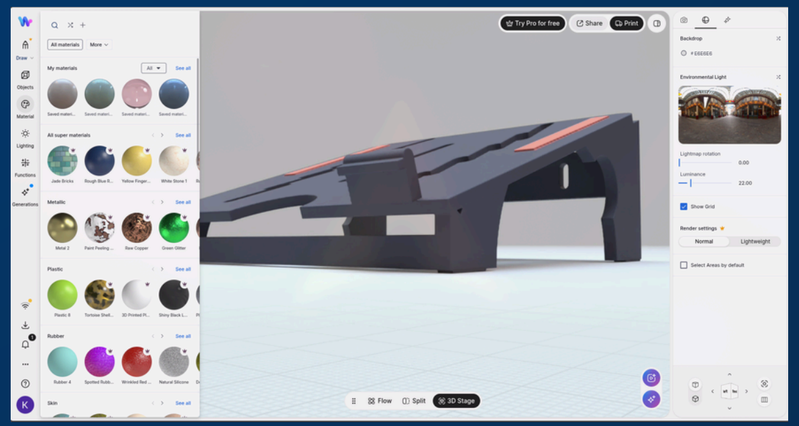
Objective and CV - Concept

Objective :

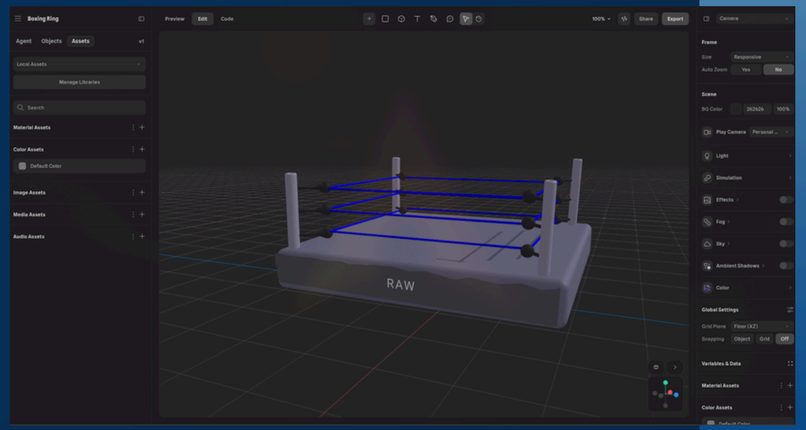
To explore Spline and Womp and understand how 3D tools can be used to represent, visualize, and interact with spatial information relevant to Computer Vision.



- Understand the purpose and capabilities of both tools
- Perform a practical 3D experiment using each tool
- Relate the experiments to 3D representation and visualization in CV



Womp



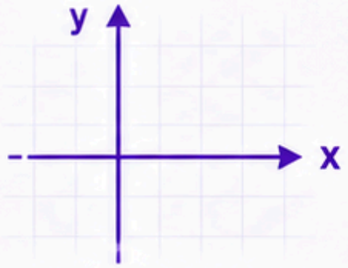
Spline

FROM 2D VISION TO 3D VISION

1

2D IMAGE

Represents information using image coordinates

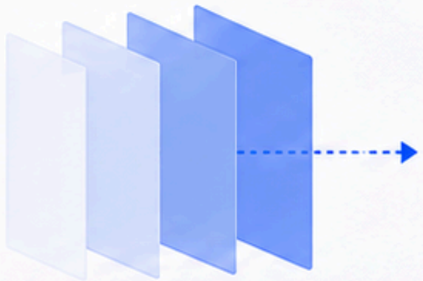


(x, y)

2

DEPTH INFORMATION

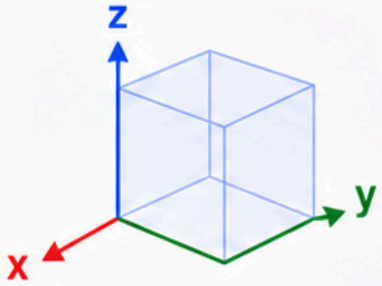
Adds depth to understand spatial structure



3

3D REPRESENTATION

Incorporates depth for true spatial understanding



(x, y, z)

3D VISION FOCUSES ON:

- Geometry
- Depth
- Spatial Relationships
- Viewpoints
- Scene Structure

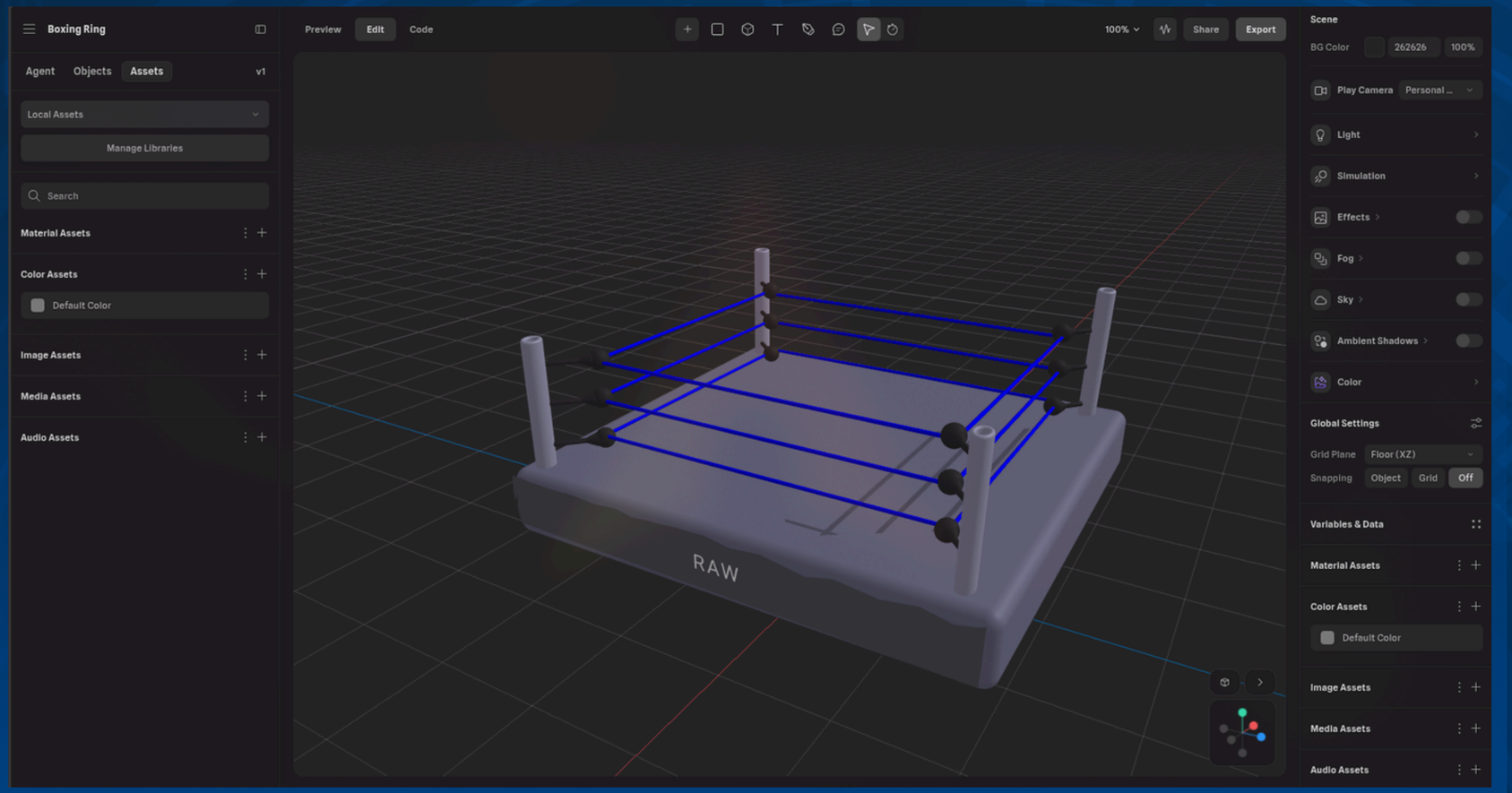
3D CV concept :

- 3D representation describes an object or scene using geometry and spatial coordinates such as X, Y, and Z.
- 3D visualization allows this representation to be viewed and explored from different viewpoints.
- This lets us reason about
  - Shape
  - Depth
  - Position
  - Orientation
  - Spatial relationships

Tool 1
Spline

What does Spline do?

- Browser-based 3D design tool
- Create and import 3D objects
- Control cameras
- Animate objects
- Add interactions
- Create interactive 3D scenes



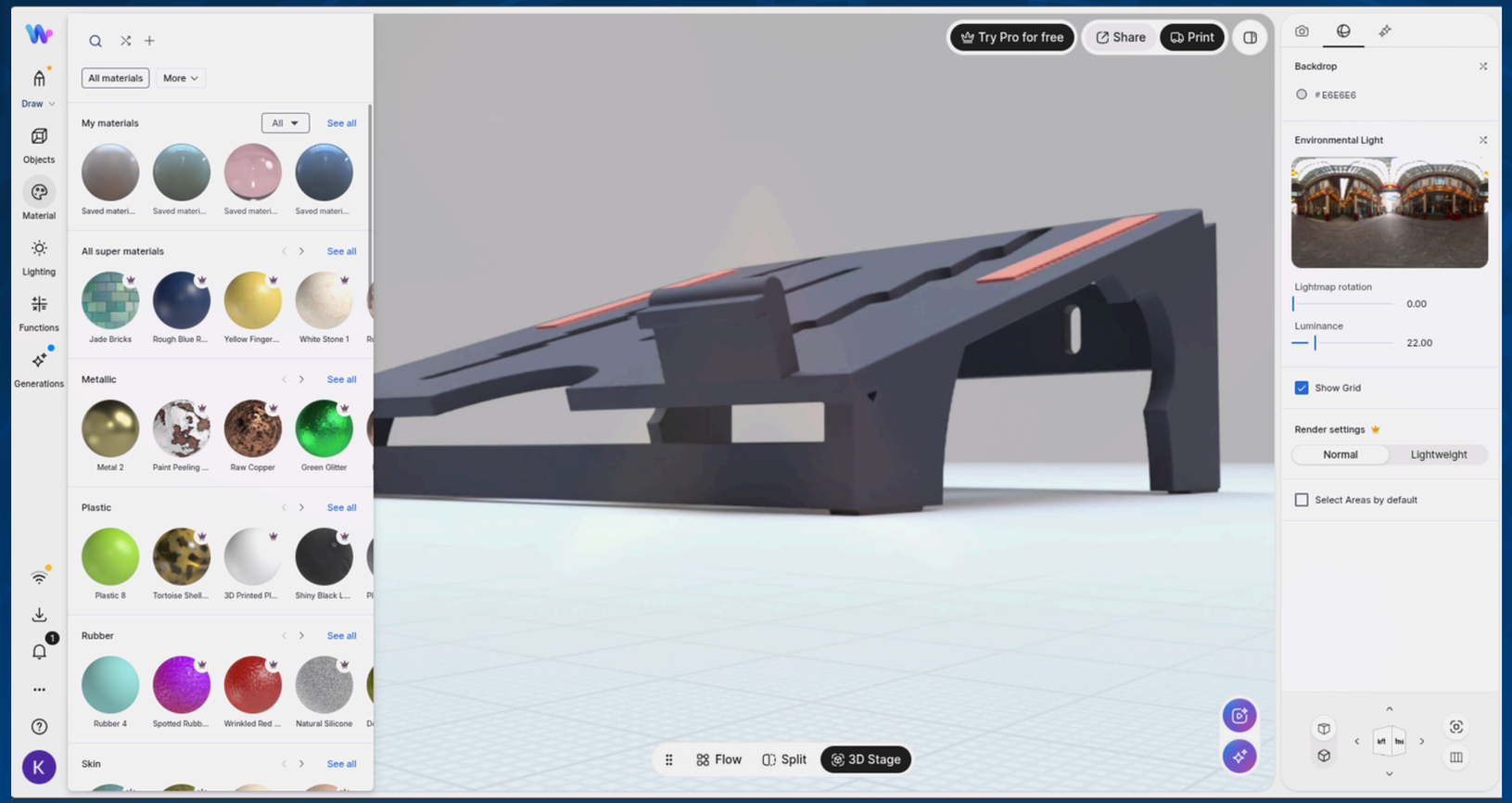
Tool Spline

Used in 3D Vision workflows to interactively visualize and explore 3D scenes, objects, and spatial information from different viewpoints.

Tool 2
Womp

What does Spline do?

- Product Design & Prototyping
- 3D Printing
- 3D Asset Creation
- Custom Products
- Character & Avatar Creation
- Architectural / Organic Design
- Image/AI → 3D Workflows



Tool Womp

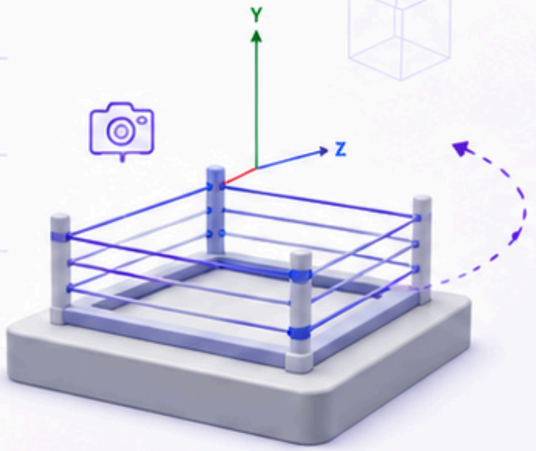
Used in 3D Vision workflows to create, modify, and refine 3D object geometry, representing the shape and structure of visual objects.

WHY SPLINE AND WOMP?

Spline

VISUALIZE • INTERACT • PRESENT

- 3D scene creation
- Camera control
- Materials and lighting
- Animation
- Interaction
- Web-based publishing



VS

womp

CREATE • MODIFY • REFINE

- 3D object modeling
- Geometric shape manipulation
- Freeform modeling
- Materials and textures
- Lighting
- Import and export

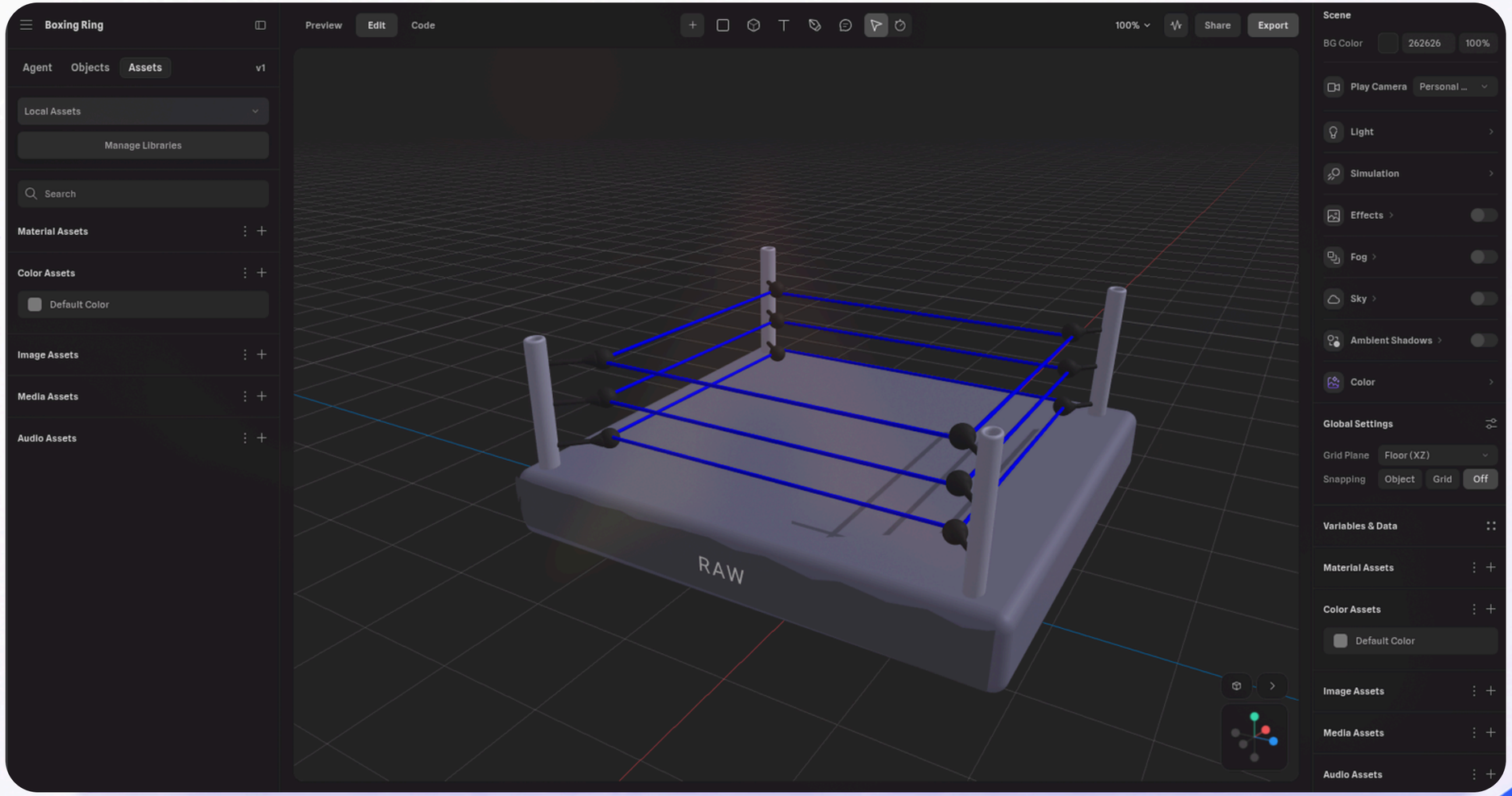
EXPERIMENT 1 — SPLINE

OBJECTIVE

Create and visualize a 3D boxing ring using basic geometric objects and scene controls.

INPUT

- Box / Platform
- Cylindrical Posts
- Rope-like Objects
- Materials
- Lighting
- Camera



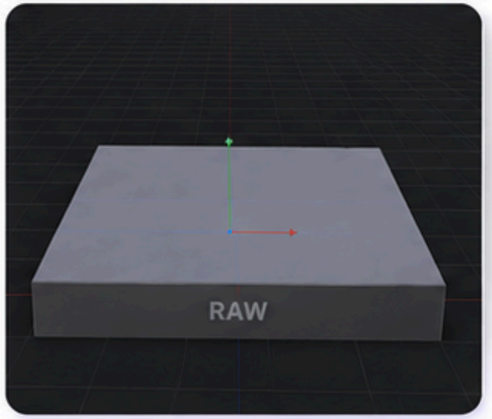
BUILDING THE 3D SCENE

Step-by-step workflow followed in Spline to create the 3D boxing ring.

01

CREATE BASE

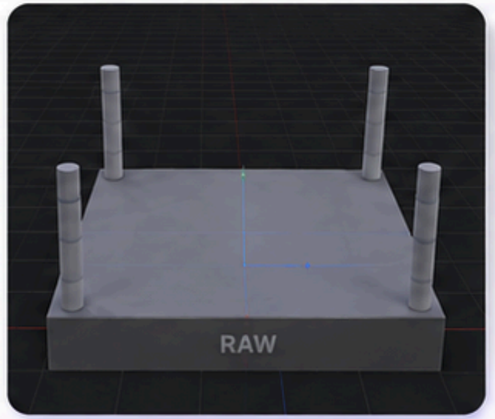
Add a box-shaped platform.



02

ADD POSTS

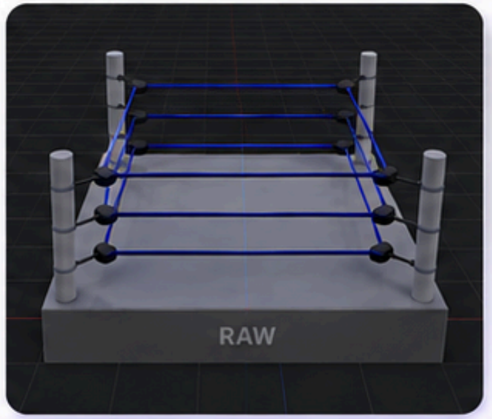
Place four cylindrical corner posts.



03

ADD ROPES

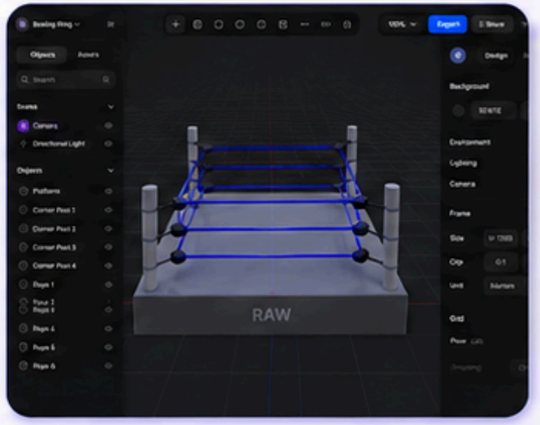
Connect posts using rope-like objects.



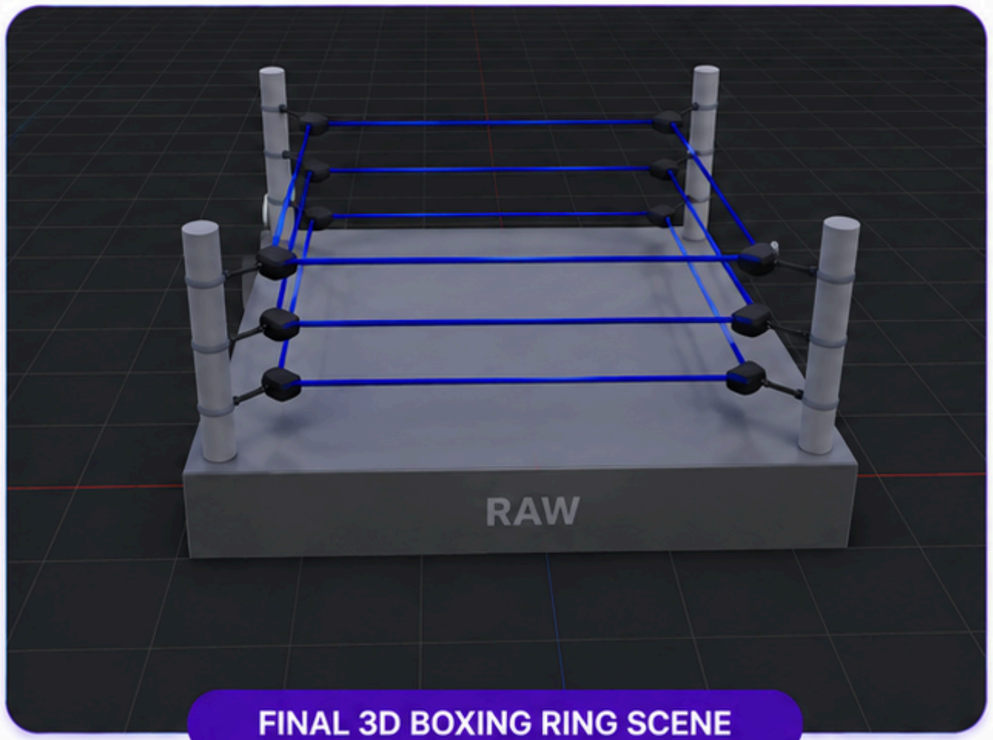
04

CONFIGURE SCENE

Adjust camera, lighting and visual settings.



SPLINE — OUTPUT & CV CONCEPT



FINAL 3D BOXING RING SCENE

KEY ELEMENTS IN THE SCENE

GEOMETRY
Platform, posts and ropes define the 3D structure.

MATERIALS
Colors and materials give visual appearance.

CAMERA
Controls viewpoint and perspective of the scene.

LIGHTING
Illumination enhances depth and scene perception.

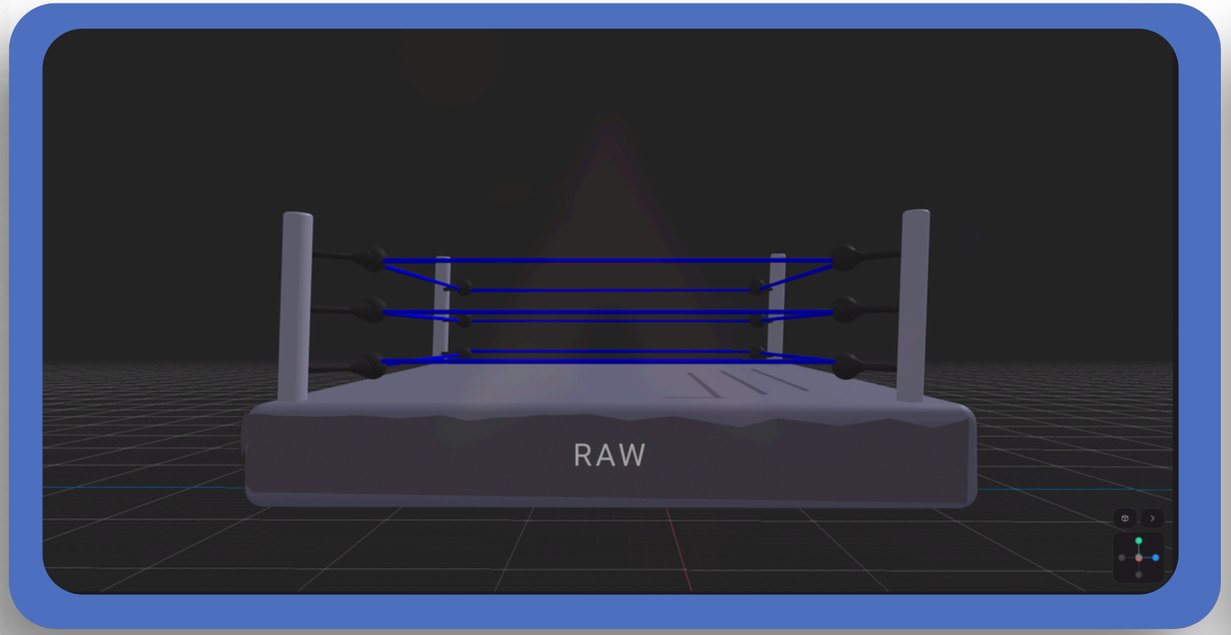
3D SCENE → CAMERA VIEWPOINT → 2D VISUAL PROJECTION

OBSERVATION
Spline makes 3D scene construction and spatial exploration intuitive through real-time visual feedback.

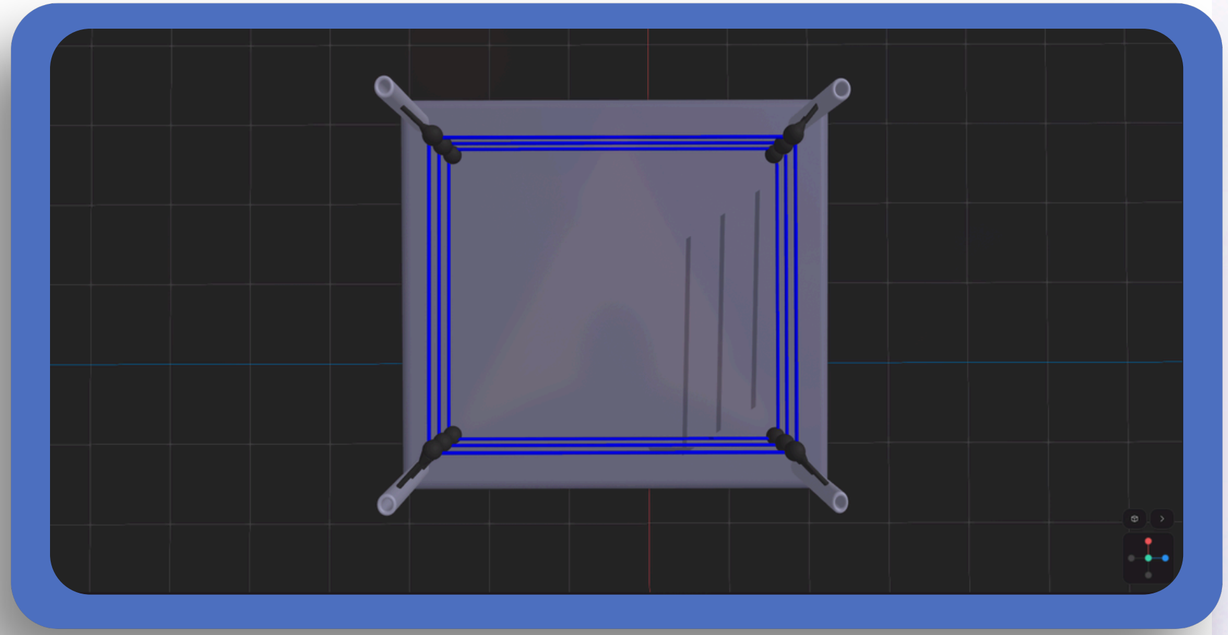
LIMITATION
It is a 3D design/visualization environment, not a dedicated CV processing framework.

BUILDING THE 3D SCENE

Step-by-step workflow followed in Spline to create the 3D boxing ring.



Front View



Top View

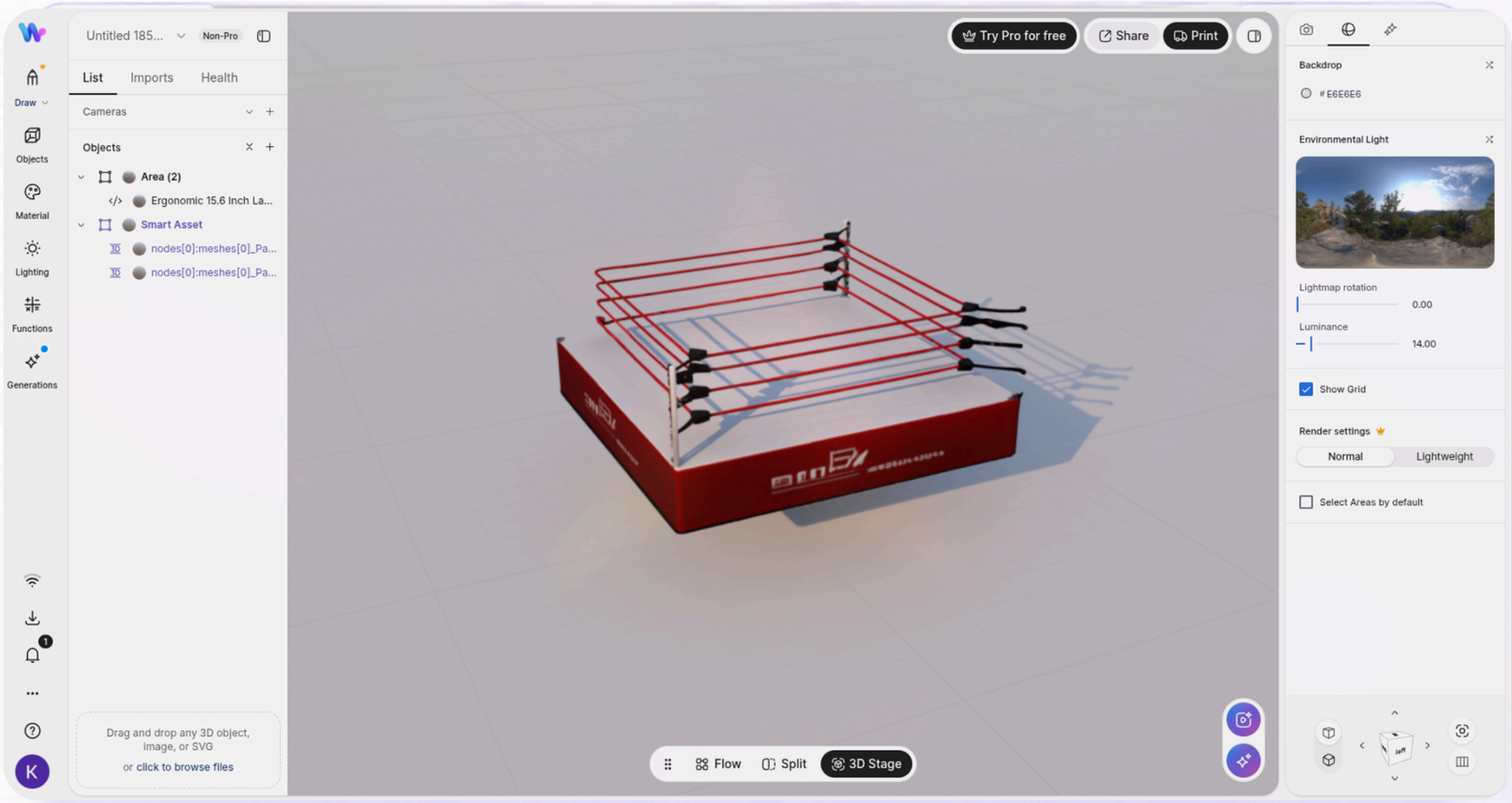
EXPERIMENT 2 — WOMP

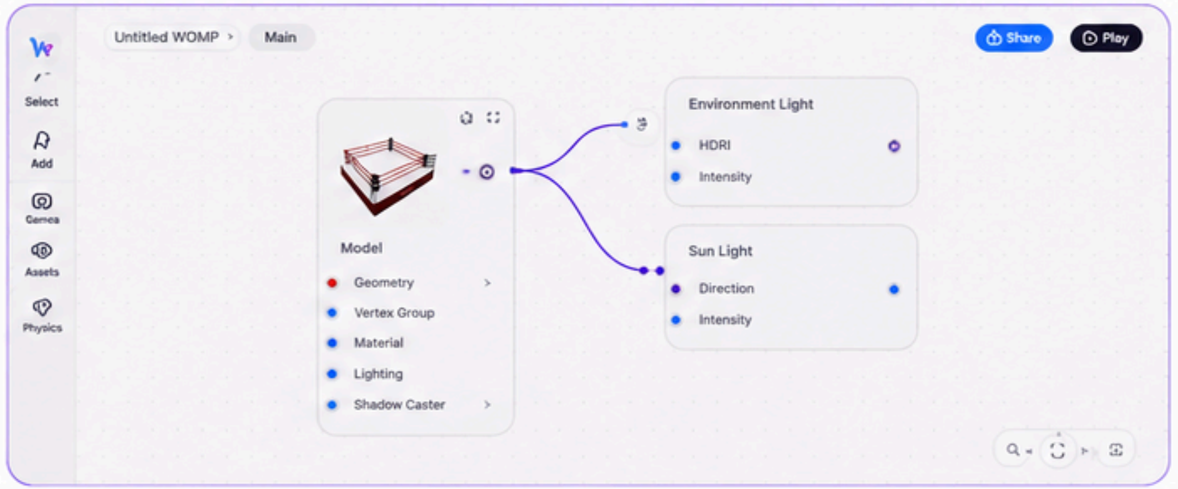
OBJECTIVE

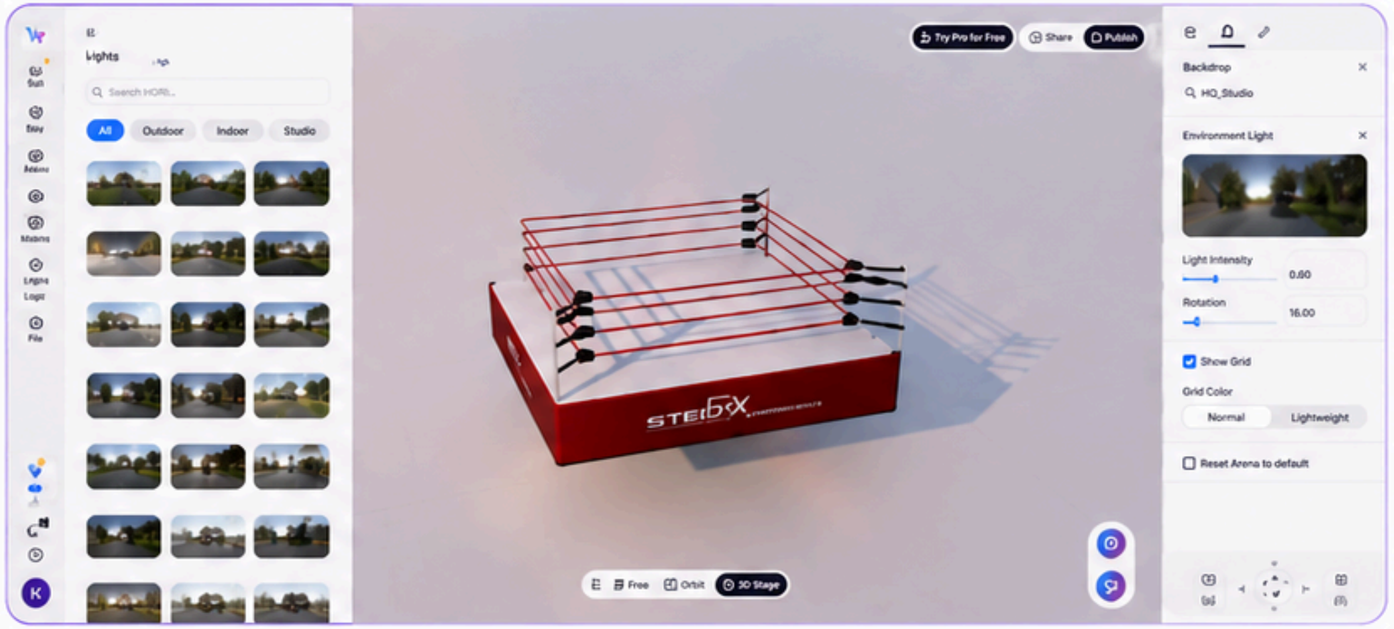
Create and refine a 3D laptop stand using geometric modeling, materials and lighting.

INPUT

- 3D Geometry
- Geometric Shapes
- Materials
- Surface Appearance
- Environmental Lighting
- Rendering Settings







3D visualization in Womp

FROM GEOMETRY TO 3D MODEL

Step-by-step workflow followed in Womp to build and refine the laptop stand.

01

CREATE

Create the object geometry and base structure.

02

MODIFY

Refine and adjust the shape for better structure.

03

APPLY

Apply materials and surface appearance.

04

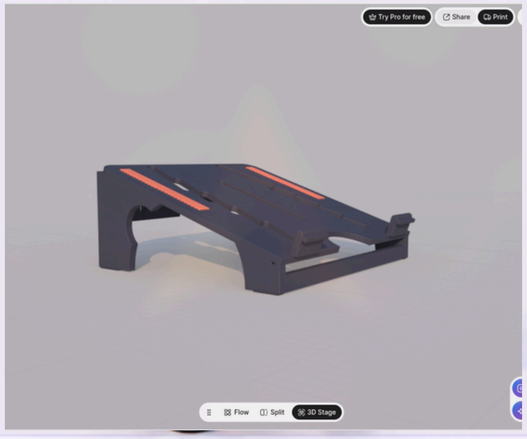
ADJUST

Adjust lighting and environment for realistic rendering.

MODEL VIEWS

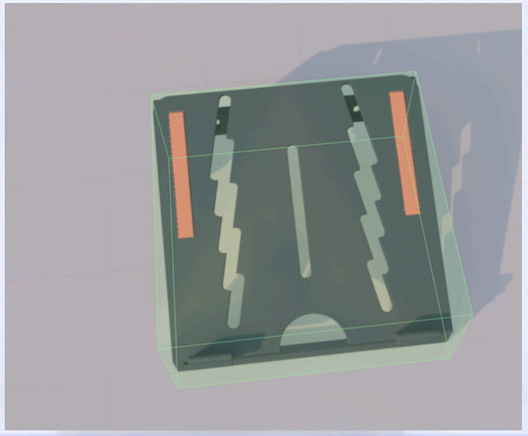
FRONT VIEW

Shows the overall front structure, slots and support features.



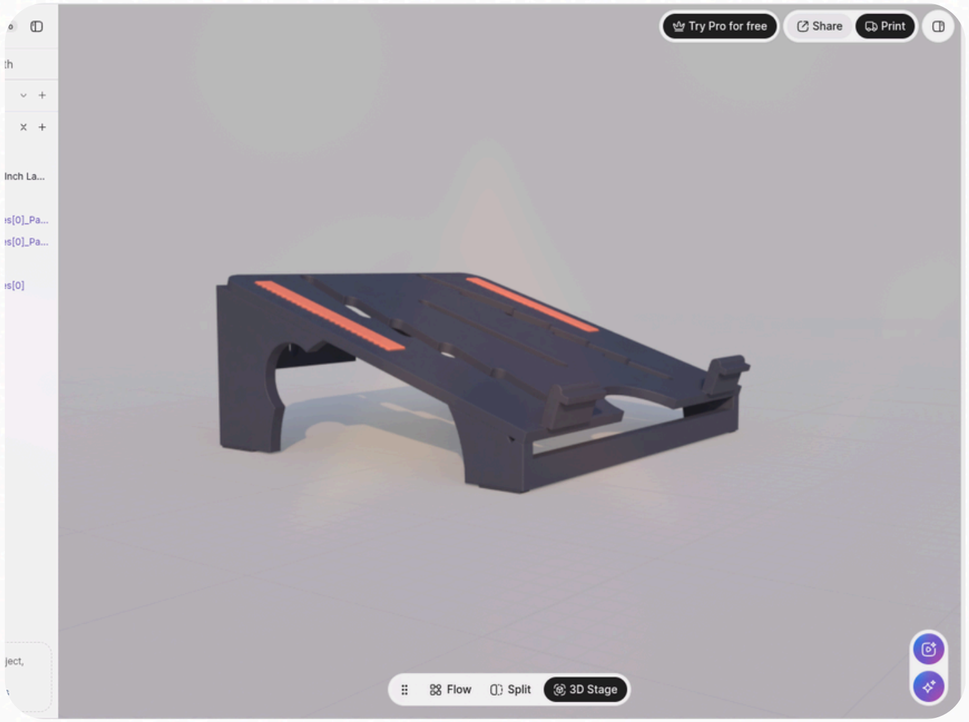
TOP VIEW

Shows the top geometry, slot placement and overall dimensions.



WOMP — 3D OBJECT REPRESENTATION

From design to fabrication-ready model.



THE 3D PRINTING WORKFLOW

GEOMETRY
Shape and spatial structure of the object.

APPEARANCE
Materials and lighting define surface look.

DIMENSIONS
Defined surfaces and object dimensions.

FABRICATION
Export & prepare for 3D printing and manufacturing.

IMPORTANT TAKEAWAY
For 3D printing, geometric correctness is more important than visual appearance alone.

KEY CONSIDERATIONS BEFORE 3D PRINTING

Dimensions
Ensure accurate measurements.

Thickness
Maintain minimum wall thickness.

Connected Geometry
Ensure all parts are properly connected.

Print Orientation
Choose optimal orientation for strength and quality.

CONCLUSION

Womp and Spline are powerful 3D tools with different strengths and complementary workflows.

WHAT WE ACHIEVED

- Created a precise, fabrication-ready 3D model in Womp.
- Exported the model using standard formats (OBJ/GLTF).
- Successfully imported the model into Spline.
- Verified geometry, materials and scene setup in Spline.

KEY TAKEAWAYS

Womp: Precision & Accuracy

Best for creating accurate, parametric and manufacturing-ready 3D models.

Spline: Creativity & Presentation

Best for interactive design, animation and visually engaging presentations.

Better Together

Womp for building it right, Spline for showing it beautifully.

FINAL TAKEAWAY

The workflow from Womp to Spline enables a smooth transition from engineering precision to creative visualization, bridging design and delivery seamlessly.

WOMP vs SPLINE — KEY DIFFERENCES

Although both tools operate in the 3D domain, their purposes and workflows are different.

Aspect	WOMP	SPLINE
Purpose	Parametric 3D modeling tool used to design and refine accurate, fabrication-ready models.	Real-time 3D design and collaboration tool used to create interactive environments, animations and presentations.
Workflow Focus	Geometry creation → refinement → materials → export	Scene building → interaction → animation → presentation/export
Modeling Approach	Parametric and precise; dimension-driven and constraint-based.	Flexible and intuitive; freeform modeling with visual feedback.
Best Suited For	Engineering designs, product models, prototypes and 3D printing.	Product showcases, UI/UX prototypes, visual storytelling and portfolios.
Output	High-accuracy, fabrication-ready 3D models (STL, OBJ, etc.).	Interactive scenes, animations, web experiences and real-time presentations.
Strength	Accuracy, parametric control and suitability for manufacturing.	Speed, real-time collaboration and engaging visual experiences.

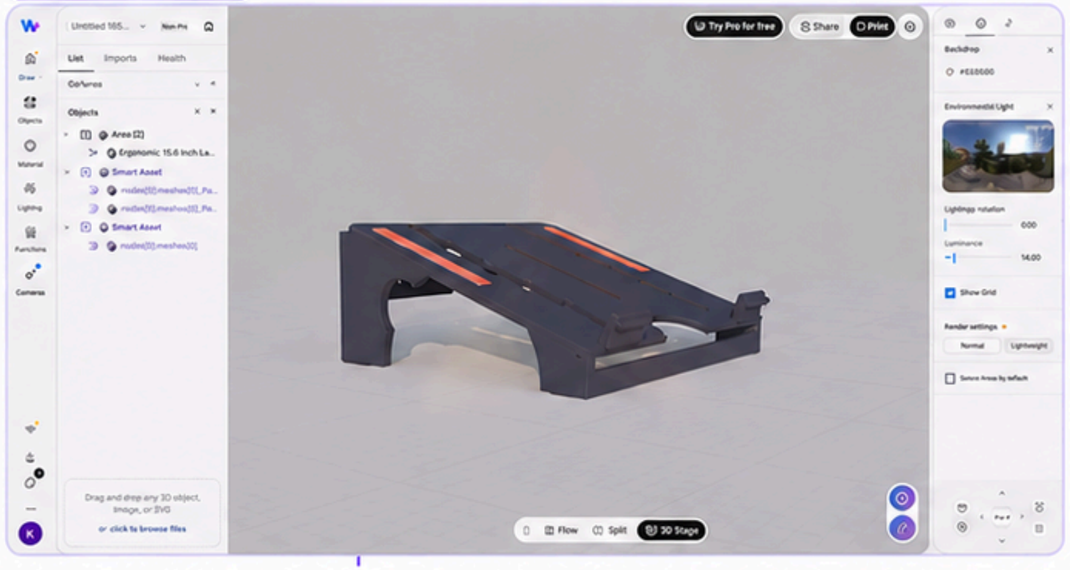
TAKEAWAY

Womp is ideal for creating precise, production-ready 3D models, while Spline excels in presenting and experiencing 3D content.

IMPORT FROM WOMP TO SPLINE

Workflow and capabilities of both tools to create a final 3D product.

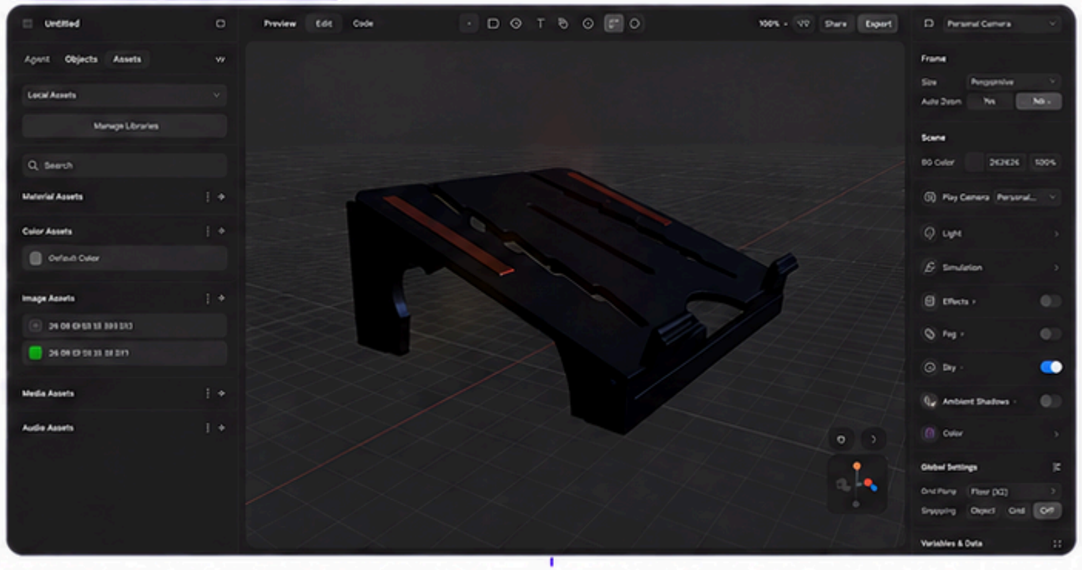
1. CREATE IN WOMP



.GLB

EXPORT (.GLB)

2. IMPORT & EDIT IN SPLINE



3. COMBINE STRENGTHS TO BUILD THE FINAL PRODUCT

WOMP — PRECISION & SETUP

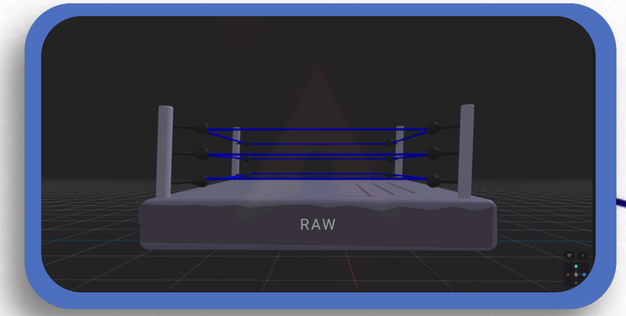
- Parametric modeling with accuracy
- Easy asset management
- Environment & lighting setup
- Export optimized 3D models

SPLINE — CREATIVITY & PRESENTATION

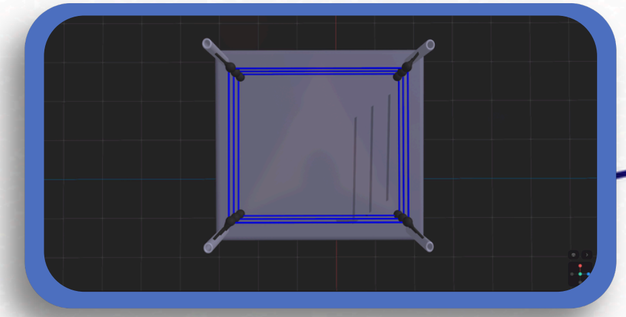
- Can be used for web UI/UX
- Lighting, effects & environment
- Animation & interaction
- High quality rendering & export

FINAL PRODUCT

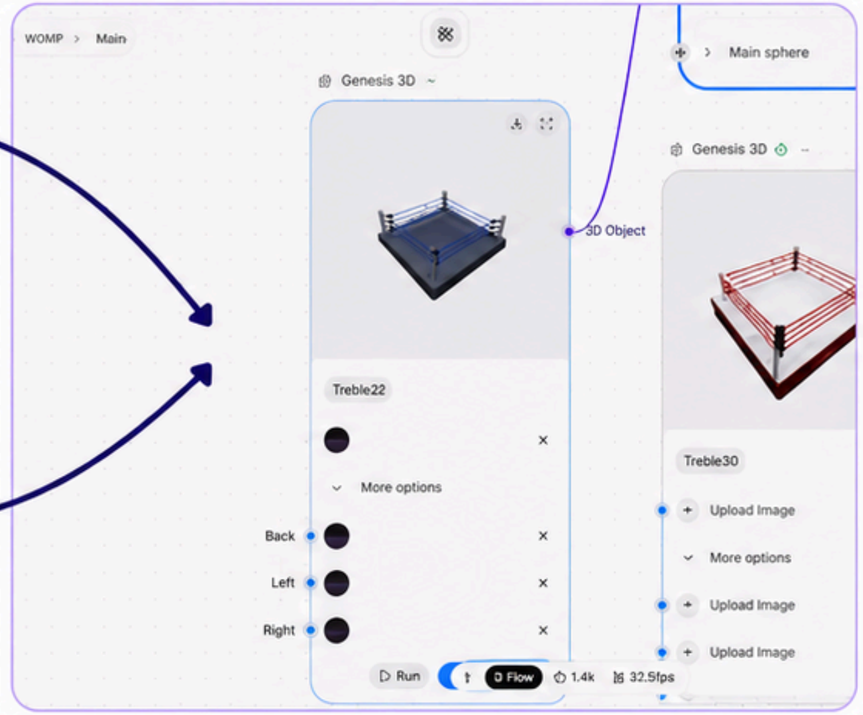
IMPORT FROM SPLINE TO WOMP



Front View



Top View



THANK YOU!

Thank you for your attention.

We hope this presentation helped you understand how Womp and Spline work together to transform ideas into accurate 3D models and engaging visual experiences.

DESIGN
with Precision

EXPORT
with Confidence

PRESENT
with Impact

From Ideas to Impact.

THANK YOU!

Questions and Discussion are Welcome.